

PROJECT

7.5

DEPLOY A WEB APP WITH CODEDEPLOY

DEVOPS SERIES

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Objective

This project automates the deployment of a web application on EC2 using AWS CodeDeploy. It integrates services like CodeBuild, CodeArtifact, and CloudFormation to streamline build and deployment, while GitHub manages version control.

IAM roles ensure secure access, and AppSpec files control the deployment process. The setup supports rollback, disaster recovery, and consistent delivery across environments.

Services

Amazon EC2 - Hosts the deployed web application.

AWS CodeDeploy - Manages the deployment process to EC2.

AWS CodeBuild - Builds and packages the application before deployment.

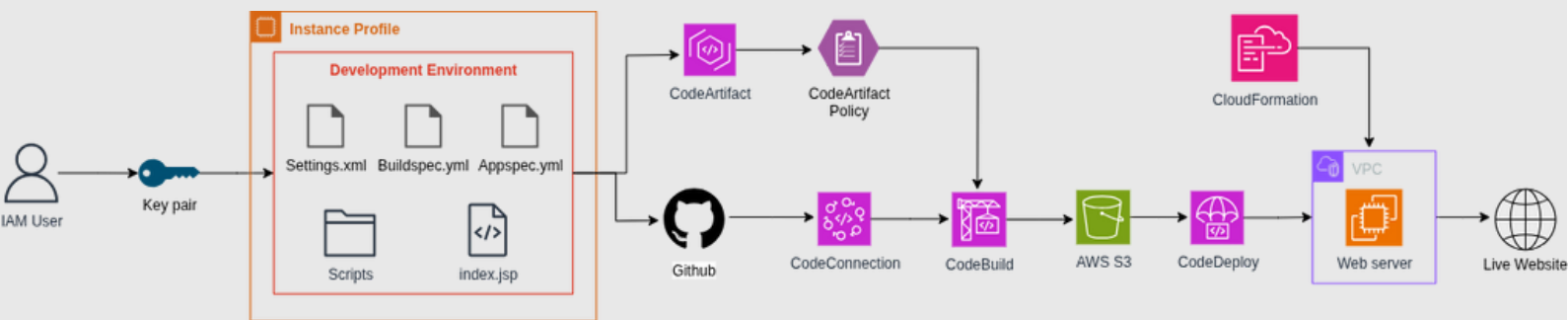
AWS CodeArtifact - Stores application dependencies and artifacts.

AWS CloudFormation - Automates EC2 infrastructure setup.

GitHub - Manages source code and triggers deployments.

AWS IAM - Provides secure access through roles and policies.

Architecture Diagram



This AWS architecture shows a CI/CD pipeline where code from GitHub is built using CodeBuild, stored in S3, and deployed with CodeDeploy to a web server in a VPC. An IAM user works in a dev environment on EC2, using CodeArtifact for dependencies and CloudFormation for infrastructure setup.

Steps to Implement

1. Launch and set up your development instance:

Start an EC2 instance, connect using VSCode Remote-SSH, and install essential tools like Git, Node.js/Java (as per your app), and the AWS CLI to prep your coding playground.

2. Set up your CodeArtifact repository:

Create a domain and repository in AWS CodeArtifact. Configure your project's settings.xml or .npmrc to securely fetch private dependencies during build time.

3. Configure CodeBuild for automated builds:

Create a CodeBuild project linked to your GitHub repo. Define a buildspec.yml to automate install, build, and packaging steps. Output artifacts are saved for deployment.

4. Launch EC2 with CloudFormation magic:

Use a CloudFormation template to automate EC2 provisioning with IAM roles, security groups, and pre-installed CodeDeploy agent for hands-free scalability.

5. Prepare your deployment scripts and AppSpec file:

Write shell scripts to start/stop the app, and create an appspec.yml to define lifecycle hooks like BeforeInstall, AfterInstall, and ApplicationStart.

6. Set up your CodeDeploy application and group:

Create a CodeDeploy application, define a deployment group, attach your EC2 instance using tags, and configure rollback settings for safe updates.

7. Trigger and verify deployment with recovery readiness:

Push your code to GitHub, let CodeDeploy handle the release, and verify it on your EC2 instance. In case of issues, auto-rollback ensures recovery using the last good revision.

Story Time

EC2 (The Cloud Castle for Your App):

EC2 is your grand castle in the cloud where your app is crafted. You launch it, equip it with tools, and access it from afar like a wizard using VSCode's Remote-SSH portal.

CodeArtifact (The Potion Vault of Dependencies):

CodeArtifact is your secure potion vault where all your project dependencies are stored. You link your app to it using special scrolls like `settings.xml` or `.npmrc`, ensuring your app always gets the right ingredients.

CodeBuild (The Magic Blacksmith):

CodeBuild is the blacksmith's workshop that automates the building of your app. It reads the `buildspec.yml` spellbook, forges your code with tests, installs, and packaging, and prepares it for battle.

CloudFormation (The Infrastructure Spellbook):

CloudFormation is the ancient spellbook that builds your entire environment in one cast. With a single YAML incantation, you summon EC2 instances, IAM roles, security groups, and install the CodeDeploy agent.

AppSpec & Scripts (The Deployment Rituals):

The `appspec.yml` and your lifecycle scripts (`start.sh`, `stop.sh`) are the sacred rituals that guide how your app is deployed. Each script runs at a specific moment to ensure a smooth and graceful deployment.

CodeDeploy (The Royal Courier):

CodeDeploy is the royal courier that delivers your newly built app to the castle (EC2). It handles deployment types, tracks progress, and ensures that if something goes wrong, a recovery spell (rollback) can be cast instantly.

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THANK YOU!

